Search in Whose Language? Separating the Query-Side Gap from the Evidence-Side Gap for Search Agents in Indic Languages

Supplementary Material

Anonymous Submission

A Comparison with the closest prior work

Table 1 lists what each close neighbour established and what our design adds.

Closest neighbour	**What it established / what it does not cover**
XBCP (Lu et al., 2026)	Deep-research agents on BrowseComp-Plus with the corpus translated into 12 languages; oracle decomposition into retrieval vs integration. Question always English; query language never logged or forced; no Indic language; translated rather than native evidence.
MarathiWeb (Kale, 2026)	Two web-search LLMs on native Marathi queries; failures attributed to integration by inspection. One language; evidence availability observed, not manipulated; no parallel English arm.
MAST Indic track (Thakur et al., 2026)	Nine Indic query languages over the English BrowseComp-Plus corpus. English-only corpus, so the evidence-side gap is zero by construction; a systems shared task.
DELTA (Park et al., 2026)	Names the gold-availability prior and regresses it out of single-shot RAG. No agent, no Indic, statistical rather than experimental control.
Where's Waldo (Ki et al., 2026)	Query-language preference with fixed paired context when a fact exists in one edition only; Bengali worst. No retrieval, no agent, no search-query language.
Diagnosing search failures (Liu et al., 2026)	Retrieval gap vs utilisation gap with oracle evidence. English only; the retrieval gap is not split into query formulation vs corpus coverage.
Cross-Lingual Cost (Amiraz et al., 2025); XRAG (Liu et al., 2025); tRAG/CrossRAG (Ranaldi, Haddow, and Birch, 2025); CORAL (Lee, Song, and Kang, 2026)	Query-language vs document-language effects in RAG; one controlled Google ablation; an agent that picks the corpus language. Non-agentic or end-accuracy only; no Indic; no decomposition.

Table 1: Novelty delineation. Comparison with the closest prior work.

B All paired contrasts

Table 2 lists every paired contrast for both languages and all four models.

Contrast, Hindi (points)	Gemini 2.5 FL	Gemini 3.1 FL	Gemini 3.6 F	Qwen3-30B
Total gap: English minus Indic question (free, both pools)	**+9.4 [+4.5, +14.3]***	+3.0 [-2.0, +8.0]	+2.0 [-4.0, +8.0]	**+9.1 [+3.8, +14.6]***
Recovered by dual-language queries	+1.8 [-3.2, +6.8]	+1.0 [-4.0, +6.0]	+0.0 [-6.0, +6.0]	+4.5 [+0.7, +8.4]*
Recovered by English-only queries	+4.5 [+0.0, +9.1]	+2.0 [-3.0, +7.0]	+6.0 [+0.0, +14.0]	-1.4 [-6.3, +3.5]
Recovered by matched-language queries	+0.7 [-3.1, +4.9]	+1.0 [-3.0, +6.1]	–	+0.3 [-2.8, +3.5]
Evidence-side: both pools minus English-only pool	-0.7 [-5.6, +3.8]	-2.0 [-7.0, +3.0]	+0.0 [-6.0, +6.0]	+5.6 [+1.0, +10.5]*
Evidence-side: both pools minus Indic-only pool	+0.3 [-3.5, +4.2]	+4.0 [-2.0, +10.1]	–	+1.4 [-1.7, +4.9]
English queries minus matched queries, English-only pool	+6.3 [+1.7, +11.2]*	+3.0 [-2.0, +8.0]	–	+3.8 [-0.7, +8.7]
English queries minus matched queries, Indic-only pool	-4.9 [-10.5, +0.3]	+0.0 [-5.0, +5.0]	–	-4.2 [-8.4, +0.0]
Symmetric monolingual: en/en minus L/L	**+10.1 [+4.5, +15.7]***	–	–	**+13.2 [+7.7, +19.2]***
Integration gap: oracle passage, English minus Indic question	**+6.3 [+2.8, +9.8]***	-2.0 [-6.0, +2.0]	-2.0 [-10.0, +4.0]	+4.2 [+0.7, +7.7]*
Closed-book gap	+3.1 [-1.0, +7.7]	+3.0 [-4.0, +10.0]	+2.0 [-4.0, +8.0]	**+12.9 [+7.3, +18.5]***
Total gap under BM25	**+18.9 [+12.9, +24.8]***	–	–	**+19.5 [+13.6, +25.8]***
Symmetric monolingual under BM25	**+35.3 [+29.0, +42.0]***	–	–	–
Recovered by dual queries under BM25	**+9.3 [+3.9, +14.6]***	–	–	**+13.6 [+8.4, +18.8]***
Indic question: BM25 minus dense (free, both)	**-15.7 [-21.3, -10.1]***	–	–	**-16.0 [-21.6, -10.5]***
Policy × evidence interaction	**+11.2 [+4.2, +18.2]***	+3.0 [-4.0, +10.0]	–	+8.0 [+2.1, +14.3]*

Table 2: Hindi questions. Paired within-item contrasts (percentage points) with item-bootstrap 95% intervals. *: sign-flip permutation $p < 0.05$. Bold: survives Benjamini–Hochberg over the full family of 144 tests (threshold $p \leq 0.0076$). Positive total gap = the English question is answered more often. Dense retriever unless marked BM25.

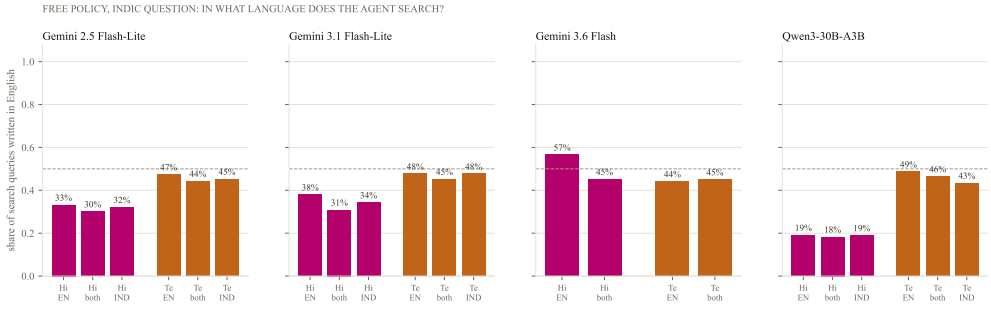


Figure 1: Share of search queries written in Latin script when the question is in Hindi (magenta) or Telugu (orange), free policy, by served pool (EN = English-only, IND = Indic-only). Dashed line: one half.

C Additional figures

D Verbatim prompts and tool schema

All prompts are reproduced exactly as sent. The policy sentence is appended to the system prompt; {`qlang`} is the name of the question's language and {`max_searches`}=3, {`k`}=5.

Agent system prompt

```
You are a search agent.  You answer factual questions by searching a document collection
with the ‘search‘ tool, reading the returned passages, and then giving a short final
answer.

Rules:

- You may call ‘search‘ at most {max_searches} times.  Each call returns the top {k}
passages.
```

Contrast, Telugu (points)	Gemini 2.5 FL	Gemini 3.1 FL	Gemini 3.6 F	Qwen3-30B
Total gap: English minus Indic question (free, both pools)	+4.7 [-1.6, +11.0]	+1.0 [-5.1, +7.1]	-6.0 [-14.0, +0.0]	+2.4 [-4.7, +9.4]
Recovered by dual-language queries	-2.5 [-9.2, +4.2]	-3.0 [-10.1, +3.0]	-2.0 [-8.0, +4.0]	+3.9 [-0.8, +9.4]
Recovered by English-only queries	+0.0 [-6.3, +6.3]	-1.0 [-6.1, +4.0]	-2.0 [-10.0, +6.0]	-1.6 [-7.1, +3.9]
Recovered by matched-language queries	-7.9 [-14.2, -1.6]*	-2.0 [-6.1, +2.0]	–	-0.8 [-6.3, +4.7]
Evidence-side: both pools minus English-only pool	-1.6 [-7.9, +4.7]	+1.0 [-3.0, +5.1]	+2.0 [-6.0, +10.0]	+2.4 [-3.1, +7.9]
Evidence-side: both pools minus Indic-only pool	+3.1 [-2.4, +8.7]	+0.0 [-6.1, +6.1]	–	+3.9 [-2.4, +11.0]
English queries minus matched queries, English-only pool	+2.4 [-3.1, +7.9]	+3.0 [-2.0, +8.0]	–	+2.4 [-4.7, +8.7]
English queries minus matched queries, Indic-only pool	+0.8 [-7.1, +8.7]	+1.0 [+0.0, +3.0]	–	-7.1 [-14.2, +0.0]
Symmetric monolingual: en/en minus L/L	+12.6 [+3.9, +21.3]*	–	–	+3.1 [-4.7, +11.0]
Integration gap: oracle passage, English minus Indic question	+6.3 [+0.0, +12.6]	-1.0 [-5.0, +2.0]	-2.0 [-6.0, +0.0]	+6.3 [+1.6, +11.0]*
Closed-book gap	+3.9 [-2.4, +11.0]	-1.0 [-7.0, +5.0]	+0.0 [-6.0, +6.0]	+7.1 [+0.0, +14.2]
Total gap under BM25	+11.0 [+2.4, +19.7]*	–	–	**+14.2 [+6.3, +22.0]***
Symmetric monolingual under BM25	**+31.0 [+20.6, +41.3]***	–	–	–
Recovered by dual queries under BM25	+7.6 [-1.7, +16.8]	–	–	+9.4 [+2.4, +17.3]*
Indic question: BM25 minus dense (free, both)	**-21.3 [-30.7, -12.6]***	–	–	**-18.9 [-28.3, -9.4]***
Policy × evidence interaction	+1.6 [-7.9, +11.0]	+2.0 [-3.0, +8.0]	–	+9.4 [+0.0, +18.9]

Table 3: Telugu questions. Paired within-item contrasts (percentage points) with item-bootstrap 95% intervals. *: sign-flip permutation $p < 0.05$. Bold: survives Benjamini–Hochberg over the full family of 144 tests (threshold $p \leq 0.0076$). Positive total gap = the English question is answered more often. Dense retriever unless marked BM25.

```
- When you have enough evidence, or when you have used all your searches, stop calling
tools and reply with the final answer only:  a short phrase (a name, date, number, or
term), no explanation.

- Answer in the same language as the question.

- If the evidence does not contain the answer, you may still answer from your own
knowledge, but prefer the evidence.{policy}
```

Policy sentence: free

```
(none)
```

Policy sentence: match

```
- Write every search query in {qlang}, the language of the question.
```

Policy sentence: english

```
- Write every search query in English, even though the question is in {qlang}.
Translate the information need into an English query.
```

Policy sentence: dual

```
- At every step call ‘search‘ twice:  once with a query in {qlang} (the language of the
question) and once with the same information need as an English query.  Both calls count
as one step.
```

Tool schema (JSON)

```
{
"type":  "function",
```

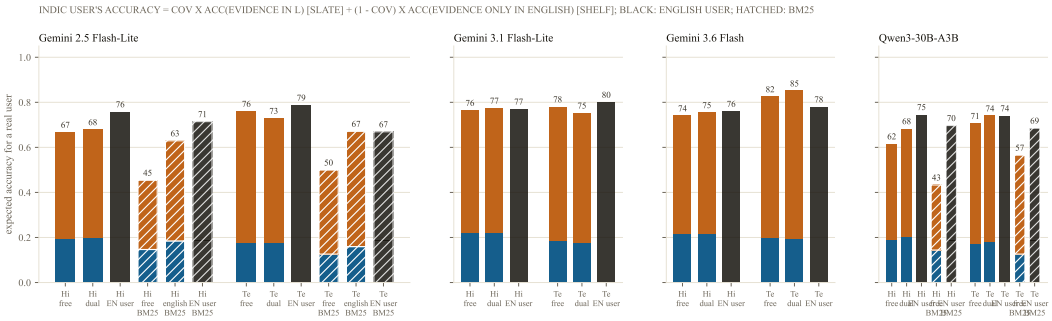


Figure 2: Expected accuracy of a real Indic user from the coverage decomposition: the slate part is coverage times accuracy with evidence in both languages; the orange part is the remaining share times accuracy on the English-only pool; black is the English user. Hatched bars use BM25.

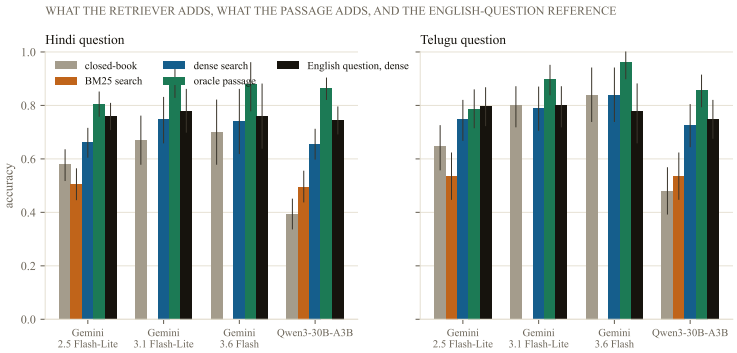


Figure 3: Indic question, free policy, both pools: closed-book, BM25 search, dense search, gold passage in the prompt, and the English question with dense search, per model.

```
"function": {
"name": "search",
"description": "Search the document collection. Returns the most relevant passages
with their titles.",
"parameters": {
"type": "object",
"properties": {
"query": {
"type": "string",
"description": "The search query."
}
},
"required": [
"query"
]
}
}
}
```

Empty-result message

```
No passages found for this query.
```

Closed-book system prompt

```
You answer factual questions from your own knowledge.  Reply with the final answer only:
a short phrase (a name, date, number, or term), no explanation.  Answer in the same
language as the question.
```

Oracle system prompt

```
You answer a factual question using the passage provided.  Reply with the final answer
only:  a short phrase (a name, date, number, or term), no explanation.  Answer in the
same language as the question.
```

Oracle user message

```
Passage:
"""
{passage}
"""
Question:  {question}
```

Judge system prompt

```
You are a strict grader of short answers to factual questions.  Decide whether the
candidate answer expresses the same fact as any of the gold answers.  Accept equivalent
forms:  a transliteration or translation of the gold answer into another language or
script, a date written differently, a number with or without units, a name with or
without honorifics, a more specific answer that contains the gold answer.  Reject
answers that name a different entity, date, or number, that hedge between several
candidates.  If the candidate declines to answer, says the information is unavailable,
or asks a question back, that is ABSTAIN.
```

Judge user message

```
Question (English):  {q_en}
Question (original):  {q_L}
Gold answers:  {gold}
Candidate answer:  {cand}
Reply with exactly one word:  CORRECT, WRONG, or ABSTAIN (the candidate says it does not
know, cannot find the answer, or asks for clarification instead of answering).
```

E Coverage measurement and audit

For each of the 539 XorQA-In test questions per language we resolved the English source article to its Hindi or Telugu counterpart through Wikidata sitelinks (916 of 918 distinct titles resolved; two had no Wikidata item) and looked the counterpart up in the 20261001 snapshot. An article exists for 364 Hindi and 283 Telugu items. Coverage was then decided in three steps. (1) *String match*: the Indic gold answer, or the English gold answer, occurs in the article after NFC normalisation and punctuation stripping; 119 Hindi and 94 Telugu items (the 237 Telugu items whose released translated answer is the function word "*ani*" were matched on the English answer only). (2) *Two independent LLM reads*: gemini-2.5-flash-lite and then gemini-3.6-flash were each given the question in both languages, the gold answers and the full article (up to 14,000 characters) and asked to quote verbatim the sentence that states the answer or reply

NONE; a read counts only if the quoted sentence is found in the article. The first read accepted 205 Hindi and 179 Telugu items; the second accepted 158 and 129; both agree on 157 and 127, which are the verified-coverage counts (29.1% and 23.6%). (3) *Manual audit*: before the second read, we drew 25 items per language stratified over first-read outcomes and labelled them by reading the quoted sentence against the question. Of the 30 first-read acceptances, 21 were correct, 7 were uncertain (the quote did not visibly contain the answer) and 2 were wrong (the Hindi article gives 13 South American countries where the gold says twelve; the Telugu Gandhi article never names Subhas Chandra Bose). The second read rejected 1 of the 2 wrong, 4 of the 7 uncertain and 8 of the 21 correct acceptances, so the final coverage figure is conservative: it under-counts rather than over-counts. The 20 first-read rejections in the audit were not labelled by hand; the second read accepted 3 of them with a verbatim quote, but an item needs both reads to count, so they stay outside the controlled grid and the coverage figure. The audit file with every label is shipped as `results/coverage_audit.json`.

F Judge audit

Of the judged agent answers, the string rule decided 41% (exact or contained match to a gold alias in either script after digit normalisation); the rest went to the LLM judge (gemini-2.5-flash, not in the grid), which returned CORRECT, WRONG or ABSTAIN. We drew 40 judged answers (20 per language: 10 string-correct, 5 LLM-correct, 5 LLM-wrong or abstain) and labelled them by hand against the question and gold aliases. All 40 verdicts agreed with the manual label. Cases worth noting: a Korean-script answer accepted for the romanised gold "mumu"; "labour movement" accepted for "trade union movement"; an answer naming one of three states in the gold marked WRONG, which we consider defensible; and seven abstentions ("the search results do not contain. . . ") marked ABSTAIN and scored as wrong. The sample is shipped as `results/audit_sample.json`.

G Per-cell tables

H Cost

I Per-cell tables

J Cost

References

Amiraz, C.; Fyodorov, Y.; Haramaty, E.; et al. 2025. The Cross-Lingual Cost: Retrieval Biases in RAG over Arabic-English Corpora. *arXiv preprint arXiv:2507.07543*.

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Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	287	57.8	[51.9, 63.4]	0%	0%
Hindi	dual	Indic-only	281	61.9	[56.2, 67.6]	50%	79%
Hindi	dual	Indic-only [bm25]	280	34.3	[28.6, 39.6]	52%	20%
Hindi	dual	both	280	67.9	[62.5, 73.6]	50%	87%
Hindi	dual	both [bm25]	281	58.7	[53.0, 64.4]	51%	67%
Hindi	dual	English-only	280	67.9	[62.1, 73.2]	50%	83%
Hindi	dual	English-only [bm25]	280	58.2	[52.5, 63.9]	51%	61%
Hindi	english	Indic-only	287	61.3	[55.7, 66.9]	100%	68%
Hindi	english	Indic-only [bm25]	285	32.3	[27.0, 37.9]	100%	8%
Hindi	english	both	287	70.7	[65.5, 76.0]	100%	80%
Hindi	english	both [bm25]	287	63.4	[57.8, 69.0]	100%	63%
Hindi	english	English-only	287	70.7	[65.5, 76.0]	100%	80%
Hindi	english	English-only [bm25]	287	62.7	[57.1, 68.3]	100%	59%
Hindi	free	Indic-only	287	65.9	[60.3, 71.4]	32%	74%
Hindi	free	Indic-only [bm25]	287	40.1	[34.5, 45.6]	38%	17%
Hindi	free	both	287	66.2	[60.6, 71.4]	30%	79%
Hindi	free	both [bm25]	286	50.3	[44.7, 56.3]	28%	40%
Hindi	free	English-only	287	66.9	[61.3, 72.5]	33%	72%
Hindi	free	English-only [bm25]	286	43.4	[37.8, 49.3]	35%	38%
Hindi	free	(gold hidden)	49	55.1	[40.8, 69.4]	27%	0%
Hindi	match	Indic-only	286	66.4	[60.8, 71.7]	1%	74%
Hindi	match	Indic-only [bm25]	286	35.3	[29.7, 40.9]	1%	16%
Hindi	match	both	287	66.9	[61.7, 72.5]	1%	78%
Hindi	match	both [bm25]	287	43.6	[38.0, 49.5]	2%	23%
Hindi	match	English-only	286	64.3	[58.7, 69.9]	1%	66%
Hindi	match	English-only [bm25]	286	19.6	[15.0, 24.5]	2%	3%
Hindi	oracle L	(gold passage)	287	72.8	[67.6, 78.0]	0%	0%
Hindi	oracle en	(gold passage)	287	80.5	[76.0, 85.0]	0%	0%
English	closed	(no tool)	287	61.0	[55.4, 66.6]	0%	0%
English	dual	Indic-only	287	64.5	[58.9, 70.0]	97%	72%
English	dual	Indic-only [bm25]	287	25.8	[20.9, 31.0]	97%	8%
English	free	Indic-only	287	65.9	[60.3, 71.1]	100%	70%
English	free	Indic-only [bm25]	286	32.2	[26.6, 37.8]	100%	8%
English	free	both	286	75.9	[71.0, 80.8]	100%	83%
English	free	both [bm25]	287	69.0	[63.4, 74.2]	100%	65%
English	free	English-only	285	75.8	[70.9, 80.7]	100%	82%
English	free	English-only [bm25]	287	71.4	[66.2, 76.7]	100%	65%
English	match	Indic-only	287	64.8	[59.2, 70.4]	100%	70%
English	match	Indic-only [bm25]	287	38.3	[32.8, 43.9]	100%	8%
English	match	both	287	74.9	[70.0, 79.8]	100%	82%
English	match	both [bm25]	287	71.4	[66.2, 76.7]	100%	65%
English	match	English-only	287	76.3	[71.4, 81.2]	100%	82%
English	match	English-only [bm25]	287	70.7	[65.5, 76.0]	100%	64%
English	oracle en	(gold passage)	287	86.8	[82.9, 90.6]	0%	0%

Table 4: All cells, Gemini 2.5 Flash-Lite, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Park, J.; Kim, B.; Hwang, S.; et al. 2026. Enhancing Multilingual RAG Systems with Debiased Language Preference-Guided Query Fusion. *arXiv preprint arXiv:2601.02956*.

Ranaldi, L.; Haddow, B.; and Birch, A. 2025. Multilingual Retrieval-Augmented Generation for Knowledge-Intensive Task. *arXiv preprint arXiv:2504.03616*.

Thakur, N.; Zhang, X.; Sharifymoghaddam, S.; Ma, X.; Chen, Z.; and Lin, J. 2026. MAST: Multilingual Agentic Search Track at FIRE 2026. `https://mast-benchmark.github.io`. Shared task; organiser baselines only as of 2026-10-10.

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	127	64.6	[55.9, 72.4]	0%	0%
Telugu	dual	Indic-only	118	71.2	[62.7, 78.8]	50%	81%
Telugu	dual	Indic-only [bm25]	118	44.9	[35.6, 54.2]	51%	13%
Telugu	dual	both	119	74.8	[66.4, 82.4]	50%	89%
Telugu	dual	both [bm25]	119	61.3	[52.9, 69.7]	50%	60%
Telugu	dual	English-only	119	72.3	[63.9, 79.8]	50%	89%
Telugu	dual	English-only [bm25]	118	54.2	[44.9, 62.7]	51%	55%
Telugu	english	Indic-only	127	67.7	[59.1, 75.6]	100%	65%
Telugu	english	Indic-only [bm25]	127	44.1	[35.4, 52.8]	100%	5%
Telugu	english	both	127	74.8	[66.9, 82.7]	100%	87%
Telugu	english	both [bm25]	127	67.7	[59.8, 75.6]	100%	62%
Telugu	english	English-only	127	77.2	[70.1, 84.3]	100%	87%
Telugu	english	English-only [bm25]	127	66.9	[58.3, 74.8]	100%	61%
Telugu	free	Indic-only	127	71.7	[63.8, 79.5]	45%	72%
Telugu	free	Indic-only [bm25]	127	43.3	[34.6, 52.0]	51%	11%
Telugu	free	both	127	74.8	[66.9, 81.9]	44%	81%
Telugu	free	both [bm25]	127	53.5	[44.9, 62.2]	43%	40%
Telugu	free	English-only	127	76.4	[68.5, 83.5]	47%	86%
Telugu	free	English-only [bm25]	127	48.8	[40.2, 57.5]	46%	43%
Telugu	free	(gold hidden)	50	72.0	[60.0, 84.0]	40%	0%
Telugu	match	Indic-only	127	66.9	[58.3, 75.6]	0%	74%
Telugu	match	Indic-only [bm25]	127	37.8	[29.1, 46.5]	2%	11%
Telugu	match	both	127	66.9	[59.1, 74.8]	0%	76%
Telugu	match	both [bm25]	127	35.4	[27.6, 44.1]	2%	14%
Telugu	match	English-only	127	74.8	[66.9, 81.9]	3%	80%
Telugu	match	English-only [bm25]	127	15.0	[9.4, 21.3]	2%	4%
Telugu	oracle L	(gold passage)	127	70.9	[63.0, 78.7]	0%	0%
Telugu	oracle en	(gold passage)	127	78.7	[71.7, 85.8]	0%	0%
English	closed	(no tool)	127	68.5	[60.6, 76.4]	0%	0%
English	dual	Indic-only	127	64.6	[55.9, 72.4]	96%	66%
English	dual	Indic-only [bm25]	127	32.3	[24.4, 40.2]	96%	9%
English	free	Indic-only	127	70.1	[62.2, 78.0]	100%	65%
English	free	Indic-only [bm25]	126	43.7	[34.9, 52.4]	100%	7%
English	free	both	127	79.5	[72.4, 86.6]	100%	88%
English	free	both [bm25]	127	64.6	[55.9, 72.4]	100%	58%
English	free	English-only	127	78.7	[71.7, 85.8]	100%	88%
English	free	English-only [bm25]	127	66.9	[58.3, 74.8]	100%	59%
English	match	Indic-only	127	70.1	[62.2, 78.0]	100%	60%
English	match	Indic-only [bm25]	126	46.0	[37.3, 54.8]	100%	6%
English	match	both	127	80.3	[73.2, 87.4]	100%	87%
English	match	both [bm25]	127	68.5	[60.6, 76.4]	100%	61%
English	match	English-only	127	79.5	[72.4, 86.6]	100%	87%
English	match	English-only [bm25]	126	69.0	[61.1, 77.0]	100%	60%
English	oracle en	(gold passage)	127	85.0	[78.7, 91.3]	0%	0%

Table 5: All cells, Gemini 2.5 Flash-Lite, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	100	67.0	[58.0, 76.0]	0%	0%
Hindi	dual	Indic-only	100	73.0	[64.0, 81.0]	51%	86%
Hindi	dual	both	100	76.0	[67.0, 84.0]	50%	90%
Hindi	dual	English-only	100	78.0	[70.0, 86.0]	50%	90%
Hindi	english	Indic-only	100	75.0	[67.0, 83.0]	100%	75%
Hindi	english	both	100	77.0	[69.0, 85.0]	100%	86%
Hindi	english	English-only	100	77.0	[68.0, 85.0]	100%	86%
Hindi	free	Indic-only	99	70.7	[61.6, 79.8]	34%	79%
Hindi	free	both	100	75.0	[66.0, 83.0]	31%	83%
Hindi	free	English-only	100	77.0	[68.0, 85.0]	38%	83%
Hindi	match	Indic-only	100	75.0	[66.0, 83.0]	5%	79%
Hindi	match	both	99	75.8	[66.7, 83.8]	7%	82%
Hindi	match	English-only	100	74.0	[65.0, 82.0]	11%	81%
Hindi	oracle L	(gold passage)	100	84.0	[76.0, 91.0]	0%	0%
Hindi	oracle en	(gold passage)	100	89.0	[83.0, 95.0]	0%	0%
English	closed	(no tool)	100	70.0	[61.0, 79.0]	0%	0%
English	free	Indic-only	100	78.0	[69.0, 86.0]	100%	76%
English	free	both	100	78.0	[70.0, 86.0]	100%	86%
English	free	English-only	100	77.0	[68.0, 85.0]	97%	84%
English	oracle en	(gold passage)	100	87.0	[80.0, 93.0]	0%	0%

Table 6: All cells, Gemini 3.1 Flash-Lite, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	100	80.0	[72.0, 87.0]	0%	0%
Telugu	dual	Indic-only	100	76.0	[67.0, 84.0]	51%	78%
Telugu	dual	both	100	76.0	[67.0, 84.0]	50%	92%
Telugu	dual	English-only	100	75.0	[66.0, 83.0]	50%	90%
Telugu	english	Indic-only	100	80.0	[72.0, 87.0]	100%	67%
Telugu	english	both	100	78.0	[69.0, 86.0]	100%	87%
Telugu	english	English-only	100	78.0	[70.0, 86.0]	100%	87%
Telugu	free	Indic-only	100	79.0	[71.0, 86.0]	48%	76%
Telugu	free	both	99	78.8	[70.7, 86.9]	45%	85%
Telugu	free	English-only	99	77.8	[69.7, 85.9]	48%	84%
Telugu	match	Indic-only	100	79.0	[71.0, 87.0]	4%	74%
Telugu	match	both	100	77.0	[69.0, 85.0]	4%	76%
Telugu	match	English-only	100	75.0	[66.0, 83.0]	6%	83%
Telugu	oracle L	(gold passage)	100	86.0	[79.0, 92.0]	0%	0%
Telugu	oracle en	(gold passage)	100	90.0	[84.0, 95.0]	0%	0%
English	closed	(no tool)	100	79.0	[71.0, 87.0]	0%	0%
English	free	Indic-only	100	79.0	[71.0, 86.0]	100%	68%
English	free	both	100	80.0	[72.0, 87.0]	100%	91%
English	free	English-only	100	80.0	[72.0, 88.0]	100%	91%
English	oracle en	(gold passage)	100	89.0	[83.0, 95.0]	0%	0%

Table 7: All cells, Gemini 3.1 Flash-Lite, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	50	70.0	[58.0, 82.0]	0%	0%
Hindi	dual	both	50	74.0	[62.0, 86.0]	50%	88%
Hindi	dual	English-only	50	76.0	[64.0, 88.0]	53%	94%
Hindi	english	both	50	80.0	[68.0, 90.0]	98%	90%
Hindi	english	English-only	50	78.0	[66.0, 88.0]	100%	90%
Hindi	free	both	50	74.0	[62.0, 86.0]	45%	80%
Hindi	free	English-only	50	74.0	[62.0, 86.0]	57%	84%
Hindi	oracle L	(gold passage)	50	78.0	[66.0, 88.0]	0%	0%
Hindi	oracle en	(gold passage)	50	88.0	[78.0, 96.0]	0%	0%
English	closed	(no tool)	50	72.0	[60.0, 84.0]	0%	0%
English	free	both	50	76.0	[64.0, 88.0]	100%	90%
English	free	English-only	50	76.0	[64.0, 88.0]	100%	84%
English	oracle en	(gold passage)	50	86.0	[76.0, 94.0]	0%	0%

Table 8: All cells, Gemini 3.6 Flash, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	50	84.0	[74.0, 94.0]	0%	0%
Telugu	dual	both	50	82.0	[70.0, 92.0]	50%	94%
Telugu	dual	English-only	50	86.0	[76.0, 94.0]	51%	94%
Telugu	english	both	50	82.0	[70.0, 92.0]	98%	94%
Telugu	english	English-only	50	76.0	[64.0, 88.0]	100%	88%
Telugu	free	both	50	84.0	[74.0, 94.0]	45%	80%
Telugu	free	English-only	50	82.0	[70.0, 92.0]	44%	92%
Telugu	oracle L	(gold passage)	50	86.0	[76.0, 94.0]	0%	0%
Telugu	oracle en	(gold passage)	50	96.0	[90.0, 100.0]	0%	0%
English	closed	(no tool)	50	84.0	[74.0, 94.0]	0%	0%
English	free	both	50	78.0	[66.0, 88.0]	100%	84%
English	free	English-only	50	78.0	[66.0, 88.0]	100%	88%
English	oracle en	(gold passage)	50	94.0	[86.0, 100.0]	0%	0%

Table 9: All cells, Gemini 3.6 Flash, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	287	39.4	[33.8, 44.9]	0%	0%
Hindi	dual	Indic-only	287	65.2	[59.6, 70.7]	50%	79%
Hindi	dual	both	287	70.0	[64.5, 75.3]	50%	87%
Hindi	dual	both [bm25]	287	63.1	[57.5, 68.6]	50%	62%
Hindi	dual	English-only	287	67.6	[62.0, 72.8]	50%	85%
Hindi	dual	English-only [bm25]	287	58.5	[53.0, 64.5]	50%	57%
Hindi	english	Indic-only	287	59.6	[54.0, 65.2]	100%	64%
Hindi	english	both	287	64.1	[58.5, 69.7]	100%	79%
Hindi	english	English-only	287	63.1	[57.5, 68.6]	100%	78%
Hindi	free	Indic-only	287	64.1	[58.5, 69.3]	19%	65%
Hindi	free	both	287	65.5	[59.9, 71.1]	18%	69%
Hindi	free	both [bm25]	287	49.5	[43.9, 55.4]	19%	23%
Hindi	free	English-only	287	59.9	[54.4, 65.5]	19%	59%
Hindi	free	English-only [bm25]	287	40.8	[35.2, 46.3]	19%	11%
Hindi	match	Indic-only	287	63.8	[58.2, 69.3]	1%	72%
Hindi	match	both	287	65.9	[60.3, 71.4]	0%	74%
Hindi	match	both [bm25]	287	43.2	[37.6, 49.1]	0%	15%
Hindi	match	English-only	287	59.2	[53.7, 64.8]	0%	65%
Hindi	match	English-only [bm25]	287	37.3	[31.7, 42.9]	0%	0%
Hindi	oracle L	(gold passage)	287	81.9	[77.4, 86.1]	0%	0%
Hindi	oracle en	(gold passage)	287	86.4	[82.2, 90.2]	0%	0%
English	closed	(no tool)	287	52.3	[46.3, 58.2]	0%	0%
English	dual	Indic-only	287	70.7	[65.2, 76.0]	100%	74%
English	free	Indic-only	287	64.8	[59.2, 70.4]	100%	55%
English	free	both	287	74.6	[69.3, 79.4]	100%	71%
English	free	both [bm25]	287	69.0	[63.4, 74.2]	100%	53%
English	free	English-only	287	74.6	[69.7, 79.4]	100%	71%
English	free	English-only [bm25]	287	69.7	[64.5, 74.9]	100%	52%
English	match	Indic-only	287	67.9	[62.4, 73.2]	100%	64%
English	match	both	287	75.6	[70.7, 80.5]	100%	80%
English	match	English-only	287	77.0	[72.1, 81.9]	100%	80%
English	oracle en	(gold passage)	287	90.6	[87.1, 93.7]	0%	0%

Table 10: All cells, Qwen3-30B-A3B, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	127	48.0	[39.4, 56.7]	0%	0%
Telugu	dual	Indic-only	127	72.4	[64.6, 80.3]	50%	80%
Telugu	dual	both	127	76.4	[69.3, 83.5]	50%	93%
Telugu	dual	both [bm25]	127	63.0	[54.3, 70.9]	50%	62%
Telugu	dual	English-only	127	73.2	[65.4, 81.1]	50%	94%
Telugu	dual	English-only [bm25]	127	65.4	[56.7, 73.2]	50%	61%
Telugu	english	Indic-only	127	66.1	[58.3, 74.0]	100%	61%
Telugu	english	both	127	70.9	[63.0, 78.7]	100%	87%
Telugu	english	English-only	127	71.7	[63.8, 79.5]	100%	87%
Telugu	free	Indic-only	127	68.5	[60.6, 76.4]	43%	61%
Telugu	free	both	127	72.4	[64.6, 80.3]	46%	70%
Telugu	free	both [bm25]	127	53.5	[44.9, 62.2]	47%	28%
Telugu	free	English-only	127	70.1	[62.2, 78.0]	49%	71%
Telugu	free	English-only [bm25]	127	57.5	[48.8, 66.1]	46%	24%
Telugu	match	Indic-only	127	73.2	[65.4, 81.1]	1%	74%
Telugu	match	both	127	71.7	[63.8, 79.5]	2%	78%
Telugu	match	both [bm25]	126	41.3	[32.5, 50.0]	1%	9%
Telugu	match	English-only	127	69.3	[61.4, 77.2]	1%	77%
Telugu	match	English-only [bm25]	127	37.8	[29.1, 46.5]	1%	1%
Telugu	oracle L	(gold passage)	127	85.0	[78.7, 91.3]	0%	0%
Telugu	oracle en	(gold passage)	127	85.8	[79.5, 91.3]	0%	0%
English	closed	(no tool)	127	55.1	[46.5, 63.8]	0%	0%
English	dual	Indic-only	127	71.7	[63.8, 79.5]	100%	71%
English	free	Indic-only	127	66.1	[57.5, 74.0]	100%	47%
English	free	both	127	74.8	[67.7, 81.9]	100%	69%
English	free	both [bm25]	127	67.7	[59.1, 75.6]	100%	42%
English	free	English-only	127	74.0	[66.1, 81.1]	100%	70%
English	free	English-only [bm25]	127	68.5	[60.6, 76.4]	100%	43%
English	match	Indic-only	127	69.3	[61.4, 77.2]	100%	57%
English	match	both	127	76.4	[68.5, 83.5]	100%	79%
English	match	English-only	127	76.4	[69.3, 83.5]	100%	83%
English	oracle en	(gold passage)	127	92.1	[87.4, 96.1]	0%	0%

Table 11: All cells, Qwen3-30B-A3B, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Stage	USD
core	9.21
uncovered	1.35
judge	1.00
coverage2	0.63
oracle	0.22
native_te	0.22
coverage	0.19
closed	0.08
noise	0.08
none	0.03
pilot	0.03
filter	0.01
Total metered (100,808 calls)	13.04

Table 12: Metered API spend (exact per-call `usage.cost`). The open-weight arm ran on a rented A100 (about 2.5 GPU-hours) and the bge-m3 embeddings took about 0.5 GPU-hour; both outside the metered total.

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	287	57.8	[51.9, 63.4]	0%	0%
Hindi	dual	Indic-only	281	61.9	[56.2, 67.6]	50%	79%
Hindi	dual	Indic-only [bm25]	280	34.3	[28.6, 39.6]	52%	20%
Hindi	dual	both	280	67.9	[62.5, 73.6]	50%	87%
Hindi	dual	both [bm25]	281	58.7	[53.0, 64.4]	51%	67%
Hindi	dual	English-only	280	67.9	[62.1, 73.2]	50%	83%
Hindi	dual	English-only [bm25]	280	58.2	[52.5, 63.9]	51%	61%
Hindi	english	Indic-only	287	61.3	[55.7, 66.9]	100%	68%
Hindi	english	Indic-only [bm25]	285	32.3	[27.0, 37.9]	100%	8%
Hindi	english	both	287	70.7	[65.5, 76.0]	100%	80%
Hindi	english	both [bm25]	287	63.4	[57.8, 69.0]	100%	63%
Hindi	english	English-only	287	70.7	[65.5, 76.0]	100%	80%
Hindi	english	English-only [bm25]	287	62.7	[57.1, 68.3]	100%	59%
Hindi	free	Indic-only	287	65.9	[60.3, 71.4]	32%	74%
Hindi	free	Indic-only [bm25]	287	40.1	[34.5, 45.6]	38%	17%
Hindi	free	both	287	66.2	[60.6, 71.4]	30%	79%
Hindi	free	both [bm25]	286	50.3	[44.7, 56.3]	28%	40%
Hindi	free	English-only	287	66.9	[61.3, 72.5]	33%	72%
Hindi	free	English-only [bm25]	286	43.4	[37.8, 49.3]	35%	38%
Hindi	free	(gold hidden)	49	55.1	[40.8, 69.4]	27%	0%
Hindi	match	Indic-only	286	66.4	[60.8, 71.7]	1%	74%
Hindi	match	Indic-only [bm25]	286	35.3	[29.7, 40.9]	1%	16%
Hindi	match	both	287	66.9	[61.7, 72.5]	1%	78%
Hindi	match	both [bm25]	287	43.6	[38.0, 49.5]	2%	23%
Hindi	match	English-only	286	64.3	[58.7, 69.9]	1%	66%
Hindi	match	English-only [bm25]	286	19.6	[15.0, 24.5]	2%	3%
Hindi	oracle L	(gold passage)	287	72.8	[67.6, 78.0]	0%	0%
Hindi	oracle en	(gold passage)	287	80.5	[76.0, 85.0]	0%	0%
English	closed	(no tool)	287	61.0	[55.4, 66.6]	0%	0%
English	dual	Indic-only	287	64.5	[58.9, 70.0]	97%	72%
English	dual	Indic-only [bm25]	287	25.8	[20.9, 31.0]	97%	8%
English	free	Indic-only	287	65.9	[60.3, 71.1]	100%	70%
English	free	Indic-only [bm25]	286	32.2	[26.6, 37.8]	100%	8%
English	free	both	286	75.9	[71.0, 80.8]	100%	83%
English	free	both [bm25]	287	69.0	[63.4, 74.2]	100%	65%
English	free	English-only	285	75.8	[70.9, 80.7]	100%	82%
English	free	English-only [bm25]	287	71.4	[66.2, 76.7]	100%	65%
English	match	Indic-only	287	64.8	[59.2, 70.4]	100%	70%
English	match	Indic-only [bm25]	287	38.3	[32.8, 43.9]	100%	8%
English	match	both	287	74.9	[70.0, 79.8]	100%	82%
English	match	both [bm25]	287	71.4	[66.2, 76.7]	100%	65%
English	match	English-only	287	76.3	[71.4, 81.2]	100%	82%
English	match	English-only [bm25]	287	70.7	[65.5, 76.0]	100%	64%
English	oracle en	(gold passage)	287	86.8	[82.9, 90.6]	0%	0%

Table 13: All cells, Gemini 2.5 Flash-Lite, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	127	64.6	[55.9, 72.4]	0%	0%
Telugu	dual	Indic-only	118	71.2	[62.7, 78.8]	50%	81%
Telugu	dual	Indic-only [bm25]	118	44.9	[35.6, 54.2]	51%	13%
Telugu	dual	both	119	74.8	[66.4, 82.4]	50%	89%
Telugu	dual	both [bm25]	119	61.3	[52.9, 69.7]	50%	60%
Telugu	dual	English-only	119	72.3	[63.9, 79.8]	50%	89%
Telugu	dual	English-only [bm25]	118	54.2	[44.9, 62.7]	51%	55%
Telugu	english	Indic-only	127	67.7	[59.1, 75.6]	100%	65%
Telugu	english	Indic-only [bm25]	127	44.1	[35.4, 52.8]	100%	5%
Telugu	english	both	127	74.8	[66.9, 82.7]	100%	87%
Telugu	english	both [bm25]	127	67.7	[59.8, 75.6]	100%	62%
Telugu	english	English-only	127	77.2	[70.1, 84.3]	100%	87%
Telugu	english	English-only [bm25]	127	66.9	[58.3, 74.8]	100%	61%
Telugu	free	Indic-only	127	71.7	[63.8, 79.5]	45%	72%
Telugu	free	Indic-only [bm25]	127	43.3	[34.6, 52.0]	51%	11%
Telugu	free	both	127	74.8	[66.9, 81.9]	44%	81%
Telugu	free	both [bm25]	127	53.5	[44.9, 62.2]	43%	40%
Telugu	free	English-only	127	76.4	[68.5, 83.5]	47%	86%
Telugu	free	English-only [bm25]	127	48.8	[40.2, 57.5]	46%	43%
Telugu	free	(gold hidden)	50	72.0	[60.0, 84.0]	40%	0%
Telugu	match	Indic-only	127	66.9	[58.3, 75.6]	0%	74%
Telugu	match	Indic-only [bm25]	127	37.8	[29.1, 46.5]	2%	11%
Telugu	match	both	127	66.9	[59.1, 74.8]	0%	76%
Telugu	match	both [bm25]	127	35.4	[27.6, 44.1]	2%	14%
Telugu	match	English-only	127	74.8	[66.9, 81.9]	3%	80%
Telugu	match	English-only [bm25]	127	15.0	[9.4, 21.3]	2%	4%
Telugu	oracle L	(gold passage)	127	70.9	[63.0, 78.7]	0%	0%
Telugu	oracle en	(gold passage)	127	78.7	[71.7, 85.8]	0%	0%
English	closed	(no tool)	127	68.5	[60.6, 76.4]	0%	0%
English	dual	Indic-only	127	64.6	[55.9, 72.4]	96%	66%
English	dual	Indic-only [bm25]	127	32.3	[24.4, 40.2]	96%	9%
English	free	Indic-only	127	70.1	[62.2, 78.0]	100%	65%
English	free	Indic-only [bm25]	126	43.7	[34.9, 52.4]	100%	7%
English	free	both	127	79.5	[72.4, 86.6]	100%	88%
English	free	both [bm25]	127	64.6	[55.9, 72.4]	100%	58%
English	free	English-only	127	78.7	[71.7, 85.8]	100%	88%
English	free	English-only [bm25]	127	66.9	[58.3, 74.8]	100%	59%
English	match	Indic-only	127	70.1	[62.2, 78.0]	100%	60%
English	match	Indic-only [bm25]	126	46.0	[37.3, 54.8]	100%	6%
English	match	both	127	80.3	[73.2, 87.4]	100%	87%
English	match	both [bm25]	127	68.5	[60.6, 76.4]	100%	61%
English	match	English-only	127	79.5	[72.4, 86.6]	100%	87%
English	match	English-only [bm25]	126	69.0	[61.1, 77.0]	100%	60%
English	oracle en	(gold passage)	127	85.0	[78.7, 91.3]	0%	0%

Table 14: All cells, Gemini 2.5 Flash-Lite, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	100	67.0	[58.0, 76.0]	0%	0%
Hindi	dual	Indic-only	100	73.0	[64.0, 81.0]	51%	86%
Hindi	dual	both	100	76.0	[67.0, 84.0]	50%	90%
Hindi	dual	English-only	100	78.0	[70.0, 86.0]	50%	90%
Hindi	english	Indic-only	100	75.0	[67.0, 83.0]	100%	75%
Hindi	english	both	100	77.0	[69.0, 85.0]	100%	86%
Hindi	english	English-only	100	77.0	[68.0, 85.0]	100%	86%
Hindi	free	Indic-only	99	70.7	[61.6, 79.8]	34%	79%
Hindi	free	both	100	75.0	[66.0, 83.0]	31%	83%
Hindi	free	English-only	100	77.0	[68.0, 85.0]	38%	83%
Hindi	match	Indic-only	100	75.0	[66.0, 83.0]	5%	79%
Hindi	match	both	99	75.8	[66.7, 83.8]	7%	82%
Hindi	match	English-only	100	74.0	[65.0, 82.0]	11%	81%
Hindi	oracle L	(gold passage)	100	84.0	[76.0, 91.0]	0%	0%
Hindi	oracle en	(gold passage)	100	89.0	[83.0, 95.0]	0%	0%
English	closed	(no tool)	100	70.0	[61.0, 79.0]	0%	0%
English	free	Indic-only	100	78.0	[69.0, 86.0]	100%	76%
English	free	both	100	78.0	[70.0, 86.0]	100%	86%
English	free	English-only	100	77.0	[68.0, 85.0]	97%	84%
English	oracle en	(gold passage)	100	87.0	[80.0, 93.0]	0%	0%

Table 15: All cells, Gemini 3.1 Flash-Lite, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	100	80.0	[72.0, 87.0]	0%	0%
Telugu	dual	Indic-only	100	76.0	[67.0, 84.0]	51%	78%
Telugu	dual	both	100	76.0	[67.0, 84.0]	50%	92%
Telugu	dual	English-only	100	75.0	[66.0, 83.0]	50%	90%
Telugu	english	Indic-only	100	80.0	[72.0, 87.0]	100%	67%
Telugu	english	both	100	78.0	[69.0, 86.0]	100%	87%
Telugu	english	English-only	100	78.0	[70.0, 86.0]	100%	87%
Telugu	free	Indic-only	100	79.0	[71.0, 86.0]	48%	76%
Telugu	free	both	99	78.8	[70.7, 86.9]	45%	85%
Telugu	free	English-only	99	77.8	[69.7, 85.9]	48%	84%
Telugu	match	Indic-only	100	79.0	[71.0, 87.0]	4%	74%
Telugu	match	both	100	77.0	[69.0, 85.0]	4%	76%
Telugu	match	English-only	100	75.0	[66.0, 83.0]	6%	83%
Telugu	oracle L	(gold passage)	100	86.0	[79.0, 92.0]	0%	0%
Telugu	oracle en	(gold passage)	100	90.0	[84.0, 95.0]	0%	0%
English	closed	(no tool)	100	79.0	[71.0, 87.0]	0%	0%
English	free	Indic-only	100	79.0	[71.0, 86.0]	100%	68%
English	free	both	100	80.0	[72.0, 87.0]	100%	91%
English	free	English-only	100	80.0	[72.0, 88.0]	100%	91%
English	oracle en	(gold passage)	100	89.0	[83.0, 95.0]	0%	0%

Table 16: All cells, Gemini 3.1 Flash-Lite, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	50	70.0	[58.0, 82.0]	0%	0%
Hindi	dual	both	50	74.0	[62.0, 86.0]	50%	88%
Hindi	dual	English-only	50	76.0	[64.0, 88.0]	53%	94%
Hindi	english	both	50	80.0	[68.0, 90.0]	98%	90%
Hindi	english	English-only	50	78.0	[66.0, 88.0]	100%	90%
Hindi	free	both	50	74.0	[62.0, 86.0]	45%	80%
Hindi	free	English-only	50	74.0	[62.0, 86.0]	57%	84%
Hindi	oracle L	(gold passage)	50	78.0	[66.0, 88.0]	0%	0%
Hindi	oracle en	(gold passage)	50	88.0	[78.0, 96.0]	0%	0%
English	closed	(no tool)	50	72.0	[60.0, 84.0]	0%	0%
English	free	both	50	76.0	[64.0, 88.0]	100%	90%
English	free	English-only	50	76.0	[64.0, 88.0]	100%	84%
English	oracle en	(gold passage)	50	86.0	[76.0, 94.0]	0%	0%

Table 17: All cells, Gemini 3.6 Flash, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	50	84.0	[74.0, 94.0]	0%	0%
Telugu	dual	both	50	82.0	[70.0, 92.0]	50%	94%
Telugu	dual	English-only	50	86.0	[76.0, 94.0]	51%	94%
Telugu	english	both	50	82.0	[70.0, 92.0]	98%	94%
Telugu	english	English-only	50	76.0	[64.0, 88.0]	100%	88%
Telugu	free	both	50	84.0	[74.0, 94.0]	45%	80%
Telugu	free	English-only	50	82.0	[70.0, 92.0]	44%	92%
Telugu	oracle L	(gold passage)	50	86.0	[76.0, 94.0]	0%	0%
Telugu	oracle en	(gold passage)	50	96.0	[90.0, 100.0]	0%	0%
English	closed	(no tool)	50	84.0	[74.0, 94.0]	0%	0%
English	free	both	50	78.0	[66.0, 88.0]	100%	84%
English	free	English-only	50	78.0	[66.0, 88.0]	100%	88%
English	oracle en	(gold passage)	50	94.0	[86.0, 100.0]	0%	0%

Table 18: All cells, Gemini 3.6 Flash, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Hindi	closed	(no tool)	287	39.4	[33.8, 44.9]	0%	0%
Hindi	dual	Indic-only	287	65.2	[59.6, 70.7]	50%	79%
Hindi	dual	both	287	70.0	[64.5, 75.3]	50%	87%
Hindi	dual	both [bm25]	287	63.1	[57.5, 68.6]	50%	62%
Hindi	dual	English-only	287	67.6	[62.0, 72.8]	50%	85%
Hindi	dual	English-only [bm25]	287	58.5	[53.0, 64.5]	50%	57%
Hindi	english	Indic-only	287	59.6	[54.0, 65.2]	100%	64%
Hindi	english	both	287	64.1	[58.5, 69.7]	100%	79%
Hindi	english	English-only	287	63.1	[57.5, 68.6]	100%	78%
Hindi	free	Indic-only	287	64.1	[58.5, 69.3]	19%	65%
Hindi	free	both	287	65.5	[59.9, 71.1]	18%	69%
Hindi	free	both [bm25]	287	49.5	[43.9, 55.4]	19%	23%
Hindi	free	English-only	287	59.9	[54.4, 65.5]	19%	59%
Hindi	free	English-only [bm25]	287	40.8	[35.2, 46.3]	19%	11%
Hindi	match	Indic-only	287	63.8	[58.2, 69.3]	1%	72%
Hindi	match	both	287	65.9	[60.3, 71.4]	0%	74%
Hindi	match	both [bm25]	287	43.2	[37.6, 49.1]	0%	15%
Hindi	match	English-only	287	59.2	[53.7, 64.8]	0%	65%
Hindi	match	English-only [bm25]	287	37.3	[31.7, 42.9]	0%	0%
Hindi	oracle L	(gold passage)	287	81.9	[77.4, 86.1]	0%	0%
Hindi	oracle en	(gold passage)	287	86.4	[82.2, 90.2]	0%	0%
English	closed	(no tool)	287	52.3	[46.3, 58.2]	0%	0%
English	dual	Indic-only	287	70.7	[65.2, 76.0]	100%	74%
English	free	Indic-only	287	64.8	[59.2, 70.4]	100%	55%
English	free	both	287	74.6	[69.3, 79.4]	100%	71%
English	free	both [bm25]	287	69.0	[63.4, 74.2]	100%	53%
English	free	English-only	287	74.6	[69.7, 79.4]	100%	71%
English	free	English-only [bm25]	287	69.7	[64.5, 74.9]	100%	52%
English	match	Indic-only	287	67.9	[62.4, 73.2]	100%	64%
English	match	both	287	75.6	[70.7, 80.5]	100%	80%
English	match	English-only	287	77.0	[72.1, 81.9]	100%	80%
English	oracle en	(gold passage)	287	90.6	[87.1, 93.7]	0%	0%

Table 19: All cells, Qwen3-30B-A3B, Hindi items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Question	Policy	Pool	n	Acc	95% CI	English queries	gold in top-5
Telugu	closed	(no tool)	127	48.0	[39.4, 56.7]	0%	0%
Telugu	dual	Indic-only	127	72.4	[64.6, 80.3]	50%	80%
Telugu	dual	both	127	76.4	[69.3, 83.5]	50%	93%
Telugu	dual	both [bm25]	127	63.0	[54.3, 70.9]	50%	62%
Telugu	dual	English-only	127	73.2	[65.4, 81.1]	50%	94%
Telugu	dual	English-only [bm25]	127	65.4	[56.7, 73.2]	50%	61%
Telugu	english	Indic-only	127	66.1	[58.3, 74.0]	100%	61%
Telugu	english	both	127	70.9	[63.0, 78.7]	100%	87%
Telugu	english	English-only	127	71.7	[63.8, 79.5]	100%	87%
Telugu	free	Indic-only	127	68.5	[60.6, 76.4]	43%	61%
Telugu	free	both	127	72.4	[64.6, 80.3]	46%	70%
Telugu	free	both [bm25]	127	53.5	[44.9, 62.2]	47%	28%
Telugu	free	English-only	127	70.1	[62.2, 78.0]	49%	71%
Telugu	free	English-only [bm25]	127	57.5	[48.8, 66.1]	46%	24%
Telugu	match	Indic-only	127	73.2	[65.4, 81.1]	1%	74%
Telugu	match	both	127	71.7	[63.8, 79.5]	2%	78%
Telugu	match	both [bm25]	126	41.3	[32.5, 50.0]	1%	9%
Telugu	match	English-only	127	69.3	[61.4, 77.2]	1%	77%
Telugu	match	English-only [bm25]	127	37.8	[29.1, 46.5]	1%	1%
Telugu	oracle L	(gold passage)	127	85.0	[78.7, 91.3]	0%	0%
Telugu	oracle en	(gold passage)	127	85.8	[79.5, 91.3]	0%	0%
English	closed	(no tool)	127	55.1	[46.5, 63.8]	0%	0%
English	dual	Indic-only	127	71.7	[63.8, 79.5]	100%	71%
English	free	Indic-only	127	66.1	[57.5, 74.0]	100%	47%
English	free	both	127	74.8	[67.7, 81.9]	100%	69%
English	free	both [bm25]	127	67.7	[59.1, 75.6]	100%	42%
English	free	English-only	127	74.0	[66.1, 81.1]	100%	70%
English	free	English-only [bm25]	127	68.5	[60.6, 76.4]	100%	43%
English	match	Indic-only	127	69.3	[61.4, 77.2]	100%	57%
English	match	both	127	76.4	[68.5, 83.5]	100%	79%
English	match	English-only	127	76.4	[69.3, 83.5]	100%	83%
English	oracle en	(gold passage)	127	92.1	[87.4, 96.1]	0%	0%

Table 20: All cells, Qwen3-30B-A3B, Telugu items (n = items; 'English queries' = share of search calls written in Latin script; 'gold in top-5' = share of episodes in which a gold chunk was retrieved).

Stage	USD
core	9.21
uncovered	1.35
judge	1.00
coverage2	0.63
oracle	0.22
native_te	0.22
coverage	0.19
closed	0.08
noise	0.08
none	0.03
pilot	0.03
filter	0.01
Total metered (100,808 calls)	13.04

Table 21: Metered API spend (exact per-call `usage.cost`). The open-weight arm ran on a rented A100 (about 2.5 GPU-hours) and the bge-m3 embeddings took about 0.5 GPU-hour; both outside the metered total.